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Game theory, compliance, and corporate criminal liability: Insights from a three-player inspection game

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ABSTRACT

The economic analysis of crime has contributed significantly to our understanding of corporate criminal activity. To derive implications for legal policy, most theoretical papers, however, take public enforcement as exogenous, i.e. non-compliance with laws is punished with a given probability by a mechanical enforcer. We apply a game-theoretic perspective to corporate criminal behavior, self-disclosure, and policing. In this paper, we model corporate compliance as a three-person inspection game in order to analyze the impact of different liability regimes on rule breaking behavior by the firm's agents, monitoring effort by corporate management and enforcement effort by authorities. Our analysis shows that one out of two equilibria in mixed strategies can be desirable for policy makers: while the first equilibrium requires a delicate regime of composite liability to induce self-disclosure, the second outcome is less prone to specification errors and does not rely on self-disclosure by firms. Which equilibrium is socially more desirable regarding social costs depends largely on the relative difference in monitoring costs between the corporation and authorities. We also find that a composite liability regime is more vulnerable to the reallocation of punishments by the agents through contracts or civil litigation. Moreover, we also consider the effects of whistleblower protection programs and a negligence-based liability regime to induce efforts to prevent crime ex-ante.

1. Introduction

Is Lady Justice "turning a blind eye to white-collar crime" [5]? Whenever the public attention is drawn to corporations that violate laws and cause public harm, the common outrage appears to fade away eventually and is followed by the impression that criminal proceedings will not be able to nail the wrongdoers. Compliance of managers and employees with laws and corporate rules is a persistent topic for lawmakers.

Since the landmark article of Becker [6], the economics of crime have contributed significantly to our understanding of (corporate) criminal activity and non-compliant behavior of employees and managers. What makes corporate crime distinct from criminal activity outside the corporation is, generally put, the involvement of three parties instead of two: the wrongdoer, the victim, and the (owner of the) corporation (see [7], p. 144–146). This is a remarkable distinction to criminal law, as firms have an advantage in monitoring their employees and revealing criminal activity compared to an outside authority (see [8,9]), given that the unlawful behavior is carried out with the company's own resources. Thus, corporations are usually capable of deterring criminal activity within their organization. Given the contractual relationship between the corporation and its employees, Garoupa [2] and other researchers interpret corporate crime as an agency cost. In this perspective, corporate criminal liability then is to mitigate the principal–agent problem and provide sufficient incentives for compliance and intra-firm policing. To achieve this, scholars have considered various legal institutions, such as strict vicarious liability and self-disclosure (see [3,7]), a composite liability regime where both employees and owners are punished (see [1,2,7]), and whistleblower protection (see [4]). A common feature of these theoretical approaches is the assumption of exogenous enforcement, i.e. the principal–agent problem in the corporation unfolds against the background of rather mechanical detection and punishment probabilities.[1]

However, Tsebelis [10] was the first to demonstrate that the focus on the agency problem of corporate compliance must not neglect the strategic interaction between the wrongdoer and the law-enforcing

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[1] In [1, p. 244, 254], the amount of harm is automatically imposed as a liability, and the authors later discuss court errors. Garoupa [2, p.6] assumes that the government will detect and punish crime with a given probability. Arlen [3, p. 844] distinguishes between two given levels of enforcement and assumes a constant detection probability. For Friebel and Guriev [4], the "punishment" is a rise or decline in the value of tradable shares, and this share price is based on rational expectations of the market.

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authority in order to derive valid policy implications. Tsebelis illustrated the limitations of the traditional approach as follows: if courts are only treated as an exogenous enforcement risk (the "black box"), then an increase in the level of sanctions increases corporate compliance. If the enforcement agency is regarded as a genuine player in the interaction, higher sanctions only reduce the frequency of law enforcement in equilibrium (see [10], p. 91). This led to the application of '*inspection games*' from game theory. An inspection game describes a strategic conflict between an inspector and an inspectee (see Fandel and Trockel 2012, p. 85). More specifically, the inspectee wants to break the law without being caught and the inspector wants compliance without costly policing. Inspection games typically assume a benevolent inspector and perfect revelation of the true behavior when the inspector decides to monitor the inspectee. Furthermore, such models usually predict that perfect deterrence cannot be achieved (see [11]).

In this paper, we want to study how different liability regimes affect corporate compliance, voluntary self-disclosure by firms and the social costs of crime when criminal behavior is modeled as the strategic interaction of three parties, the wrongdoer, the corporate management, and public authorities. Therefore, we develop a three-person inspection game and discuss the impact of different legal policy variables and institutions on equilibrium outcome and social costs.

To the best of our knowledge, the work of Fandel and Trockel [12] comes closest to our research design. In their paper, the authors describe the interaction between a manager, a controller who is to audit the manager's decisions, and the top management of the same firm. In our game, the two inspectors are of different entities (e.g., the firm's management versus authorities) and thus have different preferences. Furthermore, authorities are dependent on internal control effort by the management to reliably prosecute the illegal behavior with certainty; otherwise, the case may stay unnoticed with positive probability. Moreover, Fandel and Trockel [12, p. 92] seek to establish how equilibrium strategies are affected by bonuses and penalties while we determine total costs of crime, i.e. damage, monitoring, and error costs.

In the next chapter, we introduce the reader to the institutional background of corporate criminal liability, and present general legal considerations for imposing punishments on firms. Section 3 will then discuss the related literature. We present the setup of the inspection game in Section 4 and derive the game's equilibria in Section 5. In Section 6, we use this model to discuss corporate liability regimes and contractual reallocation of punishments. Section 7 will then provide some extensions of the model to cover further applications, such as a negligence rule and whistleblowing. Section 8 concludes.

2. The criminal liability of companies

Internationally, a threefold division can be observed with regard to corporate criminal law [13]. There are (a) countries with an independent corporate criminal law and (b) countries without such a corporate criminal law, in which only the individuals acting are subject to criminal prosecution. The former group mainly includes states with a legal system based on common law, such as the USA or Great Britain [13]. The second group traditionally includes states with civil law. However, many of these states with a codified law tradition have successively introduced corporate criminal law in recent years [14]. Finally, the third category is (c) legal systems in which corporate sanctions are not imposed under criminal law but solely under administrative law as administrative offenses [13], such as the German corporate liability for administrative offenses (§§ 30, 130 OWiG).

2.1. Statutory facts: Corporate offense

In U.S. law, the criminal liability of companies has been recognized for over 100 years. The decision "New York Central & Hudson River Rail Road Co. v. United States" of the U.S. Supreme Court in 1909 can be identified as a fundamental decision in this regard. In this landmark decision, the U.S. judiciary emancipated itself from the principle still valid at the time, namely that a company cannot act as such, cannot form a will of its own, and therefore cannot be held criminally liable itself due to a lack of culpability.

In its reasoning, the Supreme Court essentially reverted to the doctrine of civil liability. It has always been recognized that legal entities act through their organs (and thus through the people who act on their behalf) and that the consequences of these actions are therefore also attributable to them [15]. This includes both contractual law as well as tort law. In this context, the personal knowledge of acting corporate officers or representatives is deemed to be the knowledge of the organization. Commission of a tort is attributable to the represented organization [16].

The Supreme Court has transferred these principles of tort law to criminal law and developed them into the principle of vicarious criminal liability [15,17]. The prohibited act (actus reus) performed by a natural person, including the will directed toward this act (mens rea), is automatically attributable to the company if the natural person acted in the performance of his or her employee duties and thus in the area of responsibility and partly for the benefit of the company. The Anglo-Saxon concept of person facilitates this view, as it is very broadly defined and includes both natural persons and legal entities, including associations [15]. Accordingly the general criminal law applies to both natural persons and companies.

2.2. Legal consequences: Corporate sanction

The legal consequences of a criminal act provided for by law cannot be transferred congruently from natural persons to companies. This is obvious, for example, in the case of imprisonment, which is not possible for a non-natural person. Other measures from the catalog of penalties under criminal law, on the other hand, are transferable [18]. This applies first and foremost to fines. In addition, sanctions such as coercive state measures and interference in self-government autonomy are conceivable. This can mean the compulsory dismissal of corporate officers or the appointment of trustees in companies. The most severe sanction available is the forced dissolution and liquidation of the company.

In practice, as in this analysis, the focus is on fines. In the U.S., specific criteria have been developed for the assessment of fines, which are set out in the so-called Sentencing Guidelines. Formally, the sentencing courts do have discretionary powers in the assessment of fines. However, the Sentencing Guidelines clearly structure the decision-making process on sentencing, including the predictability and predictability of sentencing [19].

The actual practical significance of the Sentencing Guidelines arises for another reason. In the vast majority of cases, a court decision is not reached on the basis of a jury verdict, but rather the proceedings are suspended by negotiation with the public prosecutor's office. The Sentencing Guidelines mark the starting point for this adversarial negotiation of sanctions between the public prosecutor's office and the company. Typically, the public prosecutor's office calculates the impending penalty according to the Sentencing Guidelines as a threat and uses this as a basis for negotiating the sanction to be agreed [19].

3. Literature review

Our paper is primarily related to two major strands in the literature: corporate criminal liability and inspection games.

3.1. Corporate criminal liability

Given criminal and civil law enforcement in society, the law & economics literature has identified three major justifications for the

additional existence of corporate criminal liability. First, corporate criminal behavior entails a principal–agent problem. While the firm's employees who violate the law use resources of the corporation to obtain private gains, the violation usually does not lie in the best interest of the firm's owner (see [2,20], p. 243–245). Second, workers and managers who violate the law may often be judgment-proof to some extent as they are financially unable to bear full liability for the caused damage. This gives rise to the risk of under-deterrence and insufficient compensation for victims (see [1], p. 241). Third, the firm is an additional decision-maker, distinct from the government, who according to Arlen [7, p. 144] can help to deter crime ex-ante and ex-post. Ex-ante prevention means that the firm applies appropriate measures to prevent criminal behavior within its organization, and ex-post prevention implies monitoring and investigations to reveal wrongdoers and have them reported to authorities. In particular, the firm's ability to identify wrongdoers may be superior compared to the prosecutorial office and thus the firm can be regarded as the "least-cost enforcer" ([2], p. 244). Generally, corporate criminal liability is related to the regime of (strict) *vicarious liability*[2] in tort law. Distinct from (strict) vicarious liability in torts, however, Arlen [7, p. 145] finds that a regime that imposes strict liability on the corporation for criminal behavior of its agents cannot simultaneously induce efficient ex-ante and ex-post prevention of crime.

A crucial question in the literature is whether the peculiarities of corporate criminal law require a system of *composite liability*, i.e. a liability regime where both the firm and the individual wrongdoer receive a punishment (see [24,25], p. 434, for an overview). Garoupa [2] notes that only in a world without transaction costs and wealth constraints, where employer and employee were able to reallocate sanctions efficiently through bargaining, the question of who is punished would not matter. Garoupa further describes corporate criminal liability as a solution to a trade-off between deterrence and efficient incentives inside the firm. Polinsky and Shavell [1, p. 239-240] point out that sanctions on employees are required as firms may not be able to discipline their employees sufficiently through contractual obligations and the threat of dismissal, and the state can impose penalties in excess of what the firm can achieve through civil litigation. For Polinsky and Shavell [1, p. 251], the firm should bear the full social cost of the violation, but diminished by the amount that is transferred to the employee. Arlen [3] further identifies "conflicting, potentially perverse incentives" ([3], p. 836) of imposing strict liability on firms: as a firm that intensifies its monitoring and reporting activity increases both internal deterrence and additional litigation costs due to the firm's strict liability, it may prefer not to monitor its agents. As a consequence, Arlen proposes a reduction in fines contingent to the firm's monitoring activity, or a negligence-based liability. All these theoretical papers take public enforcement as exogenous, i.e. non-compliance with laws is punished with a given probability by a mechanical enforcer.

In addition to these general mechanisms, the literature has raised more questions and concerns on corporate criminal liability. Hamdani and Klement [26] stress that the threat to go out of business completely, even if due care by the firm is exerted, may distort (marginal) monitoring incentives. Alexander and Arlen (2018) further point out that the impact on a firm's reputation after public procedures largely depends on the future decision of parties who wish to contract with the firm. While these reputation concerns can undeniably have a strong impact on the firm's activity, such incentives cannot be controlled by the lawmaker (see [27]) and include the risk of over-deterrence.

Adding a political economy perspective, Werle [28] and Garrett [29] identify a significant reluctance by authorities to prosecute large domestic corporations. Lately, several scholars have also highlighted the role of whistleblowers (see, e.g., [30–34]) to deter corporate crime. In this regard, Mechtenberg et al. [35] find a reduced inclination of prosecutors to investigate reports but, to their surprise, no effect on deterrence when whistleblower protection is in place. For Buccirossi et al. [36], whistleblower rewards, sanctions for fraudulent reports and the standard of proof in court proceedings are intricate policy variables.

3.2. Inspection games

Distinct from decision theory approaches, inspection games take into account the strategic interaction between two decision-makers, i.e. the inspector and the inspectee. The literature on inspection games goes back to the arms race and disarmament control (see [37]) and has covered a variety of economic and political applications (for an overview, see [38]).[3] Kirstein [11, p. 386] states that an inspection game is characterized by (i) the benevolence of the inspector, (ii) her ability to perfectly monitor the inspectee but monitoring produces a cost to the inspector, and the fact that (iii) the enforcement of a punishment is costly for the inspector. Inspection games typically show a relevant Nash-equilibrium in mixed strategies and for each player, the equilibrium probability depends on the payoffs of the *other* player. In other words, the behavior of each player in equilibrium is independent of her own preferences (see [11], p. 387). In a remarkable comparison between game theory and decision theory, it was [10] who highlighted the significant implications for legal policy: if inspector and inspectee behave strategically, the claim that a higher punishment leads to less criminal behavior is no longer valid. In the world of an inspection game, the only consequence in equilibrium is the reduced probability of investigations by the inspector. This game-theoretical perspective thus offered a first indication why the commonly asserted relationship between the level of fines and the crime rate could not be substantiated by empirical researchers (see, for example, [45]). These theoretical conclusions are also in line with experimental evidence (see [46–48]), even though the author finds that the effects on observed behavior are not as strong as the theory predicts. Particularly, Rahut points out that higher fines did lead to some decline in rule-breaking behavior, and that concerning the strategic interaction, "humans adapt slowly in inspection situations" (Rahut, 2009, p. 387). To the contrary, Nosenzo et al. [49] provide experimental evidence that fines are effective and more effective in implementing compliant behavior in the inspection game than bonuses. However, this seeming disparity is simply caused by differences in modeling payoffs: if the inspection game is modeled in a zero-sum way, i.e. the inspector receives the fine as a positive gain when imposed on the inspectee, then and only then is the equilibrium rate of non-compliance contingent on the fine (as it is part of the *inspector*'s payoff).

Applying the inspection game to corporate behavior, audits and manager's incentives have been a particular focus (see, among others, [12,50,51]). In this regard, researchers have either studied the strategic interaction within the organization of the firm or extended the strategic interaction to decision-makers outside the firm. As an example of the former, Antle [52] specified a two-agent agency model and introduced a self-interested auditor into the owner-manager relationship. As the owner faces an additional principal–agent problem with the auditor, Antle describes the optimal contract to induce truthful reporting. Going one step further, Borch [53] studied the strategic interaction of a businessperson and an accountant, where the account may steal money and the businessperson can hire an auditor. The businessperson can buy insurance, but the insurer as outsider to the firm may exert effort to verify whether the businessperson fulfilled his insurance obligations and truthfully checked the accountant. Abundant applications in the corporate world exist: Siggelkow et al. [54] study the manager–auditor inspection game when both players are confronted with a whistleblower as third player. Bone and Spengler [55] shift the focus further outside the firm as they analyze corruption cases as a three-player interaction between the briber, the official and the inspector.

[2] Vicarious liability is defined as the liability of a principal for unauthorized torts of its agent when these torts occur within the scope of the principal–agent relationship (see [21], p. 1347; [22], p. 1231; [23], p. 669).

[3] Applications include, for example, fire code inspections (see [39]), tax evasion (see [40,41]), port authorities (see [42]), restaurant franchising (see [43]), and rewards for policemen (see [44]).

4. A compliance game

In the following, we develop a game-theoretic framework to study corporate criminal behavior and policing incentives under different liability regimes. Consider an inspection game with three players: the *employee (E)* of a corporation as "inspectee", the *manager (M)* of this corporation as "lower-level inspector" and *authorities (A)* as "higher-level inspector". (Corporate) laws define what kind of action is regarded as unlawful and specify fines in case of violation, which eventually lead to a punishment of non-compliant employees, corporations, or both. For the sake of simplicity of the present analysis, we ignore some of the many further principal–agent problems that may arise: we assume that the corporation's management is acting in the company's best interest and authorities are benevolent economic agents to the extent that they share society's distaste for court errors and act as impartial investigators.[4]

Inside the corporation, the employee E can commit an unobservable and illegal action that generates a private utility of b to her, but a damage to society of x with $x > b$. We assume that the social cost of crime x is large, as otherwise a policymaker would never be interested in costly criminal prosecution. Let us further specify that the illegal activity may also cause a financial loss to the corporation, which is a part of the total damage to society, and we denote the loss for the corporation through the illegal action as γx and $0 \leq \gamma < 1$.[5] Such an illegal activity could be a financial fraud where E deprives other persons of money or financial assets of value b, and the occurrence of fraud also reduces favorable transactions in the economy. Another example could be that E ignores environmental laws to her benefit, but this non-compliance generates excessive damage to the environment. Assume that if E behaves illegally and is caught by authorities, she will be subject to fine f_E with $f_E \geq b$.

The violation of legal rules can be detected inside or outside the corporation.[6]

Inside the firm, the manager M can exert costly effort with cost c_M to inspect the behavior of E and learn with certainty whether legal rules were violated.[7] However, when M does not care to inspect E and authorities learn eventually about a violation of rules by her employee, then the manager receives the fine f_M. If she did inspect and thus reported illegal behavior of E, then the fine f_M^* is imposed on the manager. This allows us to discuss different liability regimes: for instance, a strict liability would imply punishments to adhere to $f_M^* = f_M$ while a discount for self-disclosure requires $0 < f_M^* < f_M$.[8] In the basic model, we further assume that inspections by M which reveal compliant behavior by E are not prone to false accusations by a third-party and never go to court.

Outside the corporation, authorities A are the higher-level inspector of the game. If a case is indeed brought to attention, then A can inspect the case with effort costs c_A. Again, an inspection reveals the true behavior of E with certainty. If a criminal behavior is determined, then authorities rightfully impose fines upon E and M. Let A be motivated to avoid an outcome where non-compliant behavior remains unpunished.[9] Thus, in case of a correct outcome, she receives a utility of zero, and $-d$ otherwise. Let this also reflect society's preferences for correct decisions, as we assumed authorities to be benevolent in this regard. This allows us to focus on the application of internal control measures and compliance in the corporation when authorities are only motivated to enforce the law correctly. Given that the head of a public authority is usually nominated by the government, we can imagine that A's preferences coincide (to some extent) with the preferences of society. Lastly, inspections are always preferred to enforcement errors, i.e. $c_A < d$ always holds.

However, authorities A have a disadvantage in information acquisition in the field of corporate crime. We try to capture this with the following two specifications. First, authorities may hardly learn about a crime when M does not report it, and false accusations might distract authorities' attention. We thus assume that, given M does not file a report, A receives the case only with positive probability by an unknown source (e.g., the media, a whistleblower, a witness, or a victim). If E committed a crime and M did not report, then A learns of that behavior with probability $(1 - \beta)$. If E followed the rules and M did not report, A receives a (wrongful) allegation with probability α. Only then, A can inspect the situation with effort costs c_A, and potentially penalize wrongdoers. Second, we assume that it is easier for the corporation itself to identify non-compliance within the firm than it is for outsiders. To account for this disadvantage, we stylize the effort costs for inspecting the employee to be higher for authorities, thus $c_A > c_M$ applies.[10]

Furthermore, we assume that all players are risk-neutral and are inclined to maximize their expected payoffs. All this is common knowledge.

The non-cooperative game is depicted in Fig. 1 and consists of three stages: the employee decides whether to comply with the law or behave illegally at stage 1. Then, unable to observe the behavior of the employee directly, the manager as lower-level inspector can decide at stage 2 whether to examine the behavior of the employee. When the manager applies these internal control measures, the game ends, and a non-compliant employee is reported to authorities.[11] If the manager does not monitor the employee's behavior, then the case is brought to the attention of authorities with positive probability (stage 3). Authorities as higher-level inspectors decide whether to exert costly effort and examine the case. Only in the case of investigations do they impose sanctions for the revealed non-compliant behavior.

5. Equilibria and corporate liability

In this chapter, we will determine the equilibria of the three-player inspection game. To identify equilibria, we apply the concept of perfect

[4] We will thus speak of player M interchangeably as "the manager", "corporate management" or "the corporation" in this paper. We speak of "benevolent" authorities to indicate that, except for effort costs, they do not pursue a private agenda that is opposed to society's interests. Nevertheless, A is a genuine player of the game.

[5] We will also discuss the possibility of negative values of γ, i.e. the firm benefits from crime, but this seems to be a less relevant case according to the literature (see [2]). In addition to a financial loss to the firm, one could also integrate a financial damage through bad reputation whenever the criminal behavior is observed. This, however, has little effect on our model except that, if the probability of false accusations is sufficiently high, the firm will always exert monitoring effort. Note that this 'over-deterrence'-outcome is not efficient, as there will be no criminal behavior in equilibrium.

[6] Given our focus on the strategic interaction between the three players and its implications for the design of liability regimes, we follow the traditional inspection game literature here and do not employ a more refined evidence production function. Despite its simplicity, this setup still allows us to identify relevant concerns for policy makers that arise even under such idealized assumptions.

[7] Detecting non-compliance of E by own internal controls and audits can be beneficial to the company irrespective of the laws. To capture this, simply consider c_M as the net-effect of the manager's inspection effort. $c_M > 0$ must hold in our setup, otherwise 'inspection' is the dominant strategy for M, i.e. the ideal world where internal controls are always applied as the strategic dilemma of the inspection game dissolves.

[8] Such a fine reduction for self-reporting of organizations and cooperation with authorities, for example, is defined in the US Sentencing Guidelines (Section 8, 2.5g).

[9] Note that this is identical to the avoidance of decision errors in general, given that the inspection effort always reveals the true behavior of E with certainty.

[10] This follows the common notion in the literature that the corporation is regarded as "least-cost enforcer" ([2], p. 244).

[11] This implies that M has no incentive to falsely report a crime. Moreover, inspecting the employee also protects M from ungrounded prosecution in our model. We believe that applying the relevant control measures and records does significantly improve the chances of M to prove compliance to the court in the case of ungrounded real-world litigation.

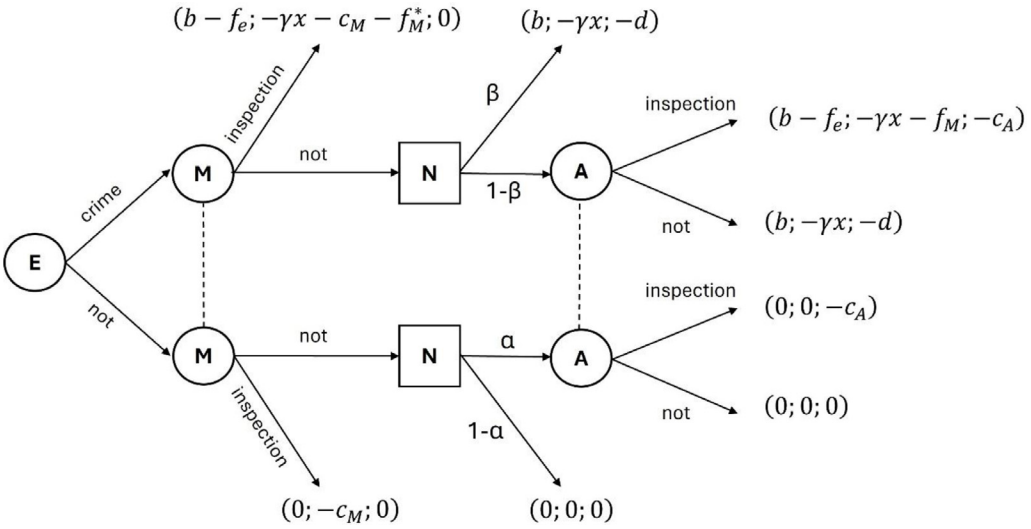


Fig. 1. Extensive form of the game.

Bayesian equilibrium (PBE) as follows: a PBE consists of the strategies $\{s_E; s_M; s_A\}$, with the strategy sets $s_E \in$ ["*crime*"; "*notcrime*"], $s_M \in$ ["*inspect*"; "*notinspect*"] and $s_A \in$ ["*inspect*"; "*notinspect*"], and the players' beliefs $\mu_M = prob$("*crime*") and $\mu_A = prob$ ("*crime*"|"*notinspect*") about the probability that the employee committed a crime "such that, at any stage of the game, strategies are optimal given the beliefs, and the beliefs are obtained from the equilibrium strategies and observed action using Bayes' rule" ([56], p. 326).

The three-player inspection game has one equilibrium in pure strategies and three equilibria in mixed strategies.[12]

In the pure-strategy equilibrium, the employee always violates the rules, M remains inactive, and authorities inspect (and punish) the employee whenever the case is brought to their attention.[13] For the non-compliant employee, this can only be favorable when the expected punishment is lower than the private benefit of crime. As authorities may not notice E's criminal behavior with probability β, the punishment f_E may be insufficient to incentivize E to follow the rules when β is large. For the corporation, "not inspect" is favorable despite the inspection by authorities if monitoring costs for the manager exceed the expected punishment discount via self-reporting, adjusted for the lower probability of trial when remaining silent and not reporting the crime.[14]

Proposition 1. *Strategies* {"*crime*"; "*notinspect*"; "*inspect*"} *and beliefs* $\mu_M = 1$ *and* $\mu_A = 1$ *constitute a PBE if* $b > (1-\beta) f_e$ *and* $c_M \geq (1-\beta) f_M - f_M^*$ *apply.*

Proof. Appendix 2.

In addition to this equilibrium in pure strategies, this inspection game shows potential equilibria in mixed strategies, i.e. when the players choose each of their pure strategies with positive probability. We thus specify as follows: E commits a crime with positive probability ϕ_E, M inspects this behavior with positive probability ϕ_M, and A inspects a case with positive probability ϕ_A (if her information set is reached). We begin with the following Corollary 1.

Corollary 1. *A mixed strategy equilibrium where either (i) M always monitors E, (ii) A never investigates the case or (iii) E chooses a pure strategy cannot exist.*

Proof. Obvious from the text, therefore omitted.

The reasoning is straightforward: any mixed strategy equilibrium requires that at least two players randomize between their pure strategies, i.e. no more than one player can stick to a pure strategy. (i) When M plays the pure strategy "*inspect*", E receives a certain punishment when choosing "*crime*". Given $f_E \geq b$ holds by assumption, she will play the pure strategy "*no crime*", and a mixed strategy outcome cannot exist. (ii) If A never investigates a case, M will never be punished except due to voluntary self-disclosure. Thus, M will strictly prefer the costless pure strategy "*not inspect*", and again at least two players do not randomize between their pure strategies. (iii) If E always chooses to comply with the rules, neither A nor M are inclined to inspect the case. Also note from Fig. 1 that the behavior of M does not affect the belief of A about the probability of crime, i.e. M cannot make authorities indifferent between their pure strategies. Thus, if the pure strategy "*crime*" is chosen in equilibrium, then A cannot be indifferent, and Proposition 1 must apply. We now turn to the conditions that make each player randomize.

Authorities form rational beliefs μ_A about the probability that E chose "*crime*" whenever their decision nod is reached. Then, according to Bayes' rule, A will believe that the employee violated the law with

$$\mu_A = \frac{\phi_E(1-\beta)}{\phi_E(1-\beta) + (1-\phi_E)\alpha} \tag{1}$$

Furthermore, authorities will be indifferent between "*inspect*" and "*not inspect*" at their decision nod if the condition $-c_A = -\mu_A d$ holds for belief μ_A. Inserting (1) and solving for ϕ_E defines the probability of E choosing "*crime*" to make A indifferent in equilibrium, and this gives

$$\phi_E^* = \frac{\alpha c_A}{(1-\beta)d - (1-\alpha-\beta)c_A} \tag{2}$$

In other words, Eq. (2) yields the probability of crime in equilibrium.[15] Note that, as M is unable to affect the updated belief μ_A of authorities in our game, the behavior of the employee in equilibrium is not dependent on the probabilities that M or A choose to inspect her behavior. The probability of "*crime*" increases when inspection costs

[12] See Appendix 1 for the normal form of the 2 × 2 × 2 game.

[13] A pure strategy equilibrium where the employee always violates the rule, and the manager always reports this behavior is excluded by assumption of condition $f_E > b$.

[14] In addition to excessive monitoring costs, this outcome may be implemented by the firm if we considered the crime to be sufficiently profitable to the firm.

[15] The probability is between zero and one as $d > c_A$ always holds in our model.

are higher for authorities or when more false allegations are reported and decreases with the authority's sensitivity to enforcement errors and when more cases are reported correctly to authorities.

The manager also forms beliefs μ_M about the probability that E chose "*crime*", and rationality clearly implies $\mu_M = \phi_E$. Furthermore, M is made indifferent between "*inspect*" and "*not inspect*" if the probability of committing a crime and the probability of inspections by authorities enable $\mu_M\left(\gamma x + c_M + f_M^*\right) + \left(1-\mu_M\right)c_M = \mu_M\left(\gamma x + \phi_A(1-\beta)f_M\right)$ to hold. Using $\mu_M = \phi_E$ and solving for ϕ_A, we find the probability of inspections by authorities that turn the manager indifferent, as

$$\phi_A^*(\phi_E) = \frac{\phi_E f_M^* + c_M}{\phi_E(1-\beta)f_M} \tag{3}$$

Evidently, authorities will have to inspect more cases in equilibrium if monitoring costs for the manager are high and sanction discounts for monitoring managers are low (i.e., f_M^* is high and close to f_M). A higher probability of observing unreported cases and higher sanctions for careless managers reduce the required inspection effort by A. Furthermore, the equilibrium probability is between zero and one, if and only if

$$c_M < \phi_E\left[(1-\beta)f_M - f_M^*\right]. \tag{4}$$

Condition (4), a boundary condition for the later equilibrium, is reminiscent of the requirement for the pure strategy outcome: the manager will have only an incentive to monitor her employee if the (expected) benefit of reporting a non-compliant employee exceeds certain effort costs. As crime only occurs with positive probability, this condition is more likely to hold when the crime rate ϕ_E is higher. Otherwise, managers always stay silent.

Lastly, the employee is made indifferent between "*crime*" or "*no crime*" if both inspectors randomize between monitoring her behavior and applying no control effort, as $\phi_M\left(b - f_e\right) + \left(1-\phi_M\right)\phi_A\left(b-(1-\beta)f_e\right) + \left(1-\phi_M\right)\left(1-\phi_A\right)b = 0$ holds. Thus, the combination of monitoring probabilities by the two inspectors, ϕ_M and ϕ_A, also has to satisfy this condition in equilibrium. Rearranging for ϕ_M, we find E to be indifferent for

$$\phi_M = \frac{b - \phi_A(1-\beta)f_E}{f_E - \phi_A(1-\beta)f_E} \tag{5}$$

Given the probability of internal inspections by M, ϕ_M, we can derive a second requirement for the three-player mixed-strategy outcome: For (5) to be between zero and one, a non-negative numerator requires the condition

$$b > \phi_A(1-\beta)f_E \tag{6}$$

to hold. This implies that the employee would always choose the pure strategy "*crime*" if only authorities inspected the behavior with positive probability, ϕ_A, and the expected punishment were too low compared to the private gains. Consequently, $\phi_M > 0$ has to apply in equilibrium when (6) holds.

Using conditions (3) and (5), we can now determine the probability that the manager monitors her employee in equilibrium as

$$\phi_M = \frac{\phi_E\left(bf_M - f_E f_M^*\right) - f_E c_M}{\phi_E\left(f_E f_M - f_E f_M^*\right) - f_E c_M} \tag{7}$$

Note that the derived solution for ϕ_M can never equal one given $b < f_E$. Furthermore, the numerator is positive if only $c_M < \phi_E\left[\frac{b}{f_E}f_M - f_M^*\right]$ holds.[16] Given these considerations, we can identify the following mixed strategy outcomes: an equilibrium where all three players randomize, and two equilibria in partially mixed strategies where the employer and either authorities or the manager randomize, and the third player chooses a pure strategy. This leads to Proposition 2.

[16] This also follows from inserting condition (3) into inequality (6).

Proposition 2. *The following mixed strategy equilibria exist:*

(2.1) $\phi_E^{2.1} = \frac{\alpha c_A}{(1-\beta)d-(1-\alpha-\beta)c_A}$, $\phi_M^{2.1} = \frac{\phi_E\left(bf_M - f_E f_M^*\right) - f_E c_M}{\phi_E\left(f_E f_M - f_E f_M^*\right) - f_E c_M}$ *and* $\phi_A^{2.1} = \frac{\phi_E f_M^* + c_M}{\phi_E(1-\beta)f_M}$, *given beliefs* $\mu_M = \phi_E$ *and* $\mu_A = \frac{c_A}{d}$. *This requires conditions* $c_M < \phi_E\left[(1-\beta)f_M - f_M^*\right]$ *and* $c_M < \phi_E\left[\frac{b}{f_E}f_M - f_M^*\right]$.

(2.2) $\phi_E^{2.2} = \frac{\alpha c_A}{(1-\beta)d-(1-\alpha-\beta)c_A}$, $\phi_M^{2.2} = 0$ *and* $\phi_A^{2.2} = \frac{b}{(1-\beta)f_E}$, *given beliefs* $\mu_M = \phi_E$ *and* $\mu_A = \frac{c_A}{d}$. *This requires the conditions* $b < (1-\beta)f_E$ *and* $c_M > \phi_E\left[\frac{b}{f_E}f_M - f_M^*\right]$ *to hold.*

(2.3) $\phi_E^{2.3} = \frac{c_M}{(1-\beta)f_M - f_M^*}$, $\phi_M^{2.3} = \frac{b-(1-\beta)f_E}{\beta f_E}$ *and* $\phi_A^{2.3} = 1$, *given beliefs* $\mu_M = \phi_E$ *and* $\mu_A > \frac{c_A}{d}$. *This requires* $b > (1-\beta)f_E$, $c_M < (1-\beta)f_M - f_M^*$ *and* $\phi_E^{2.3} > \phi_E^{2.1}$ *holds.*

Proof. Appendix 2.

Having identified three different equilibria, a few remarks are in order.

First, the three equilibria clearly have different properties regarding efficiency, and particularly, deterrence. Following our initial assumption that the social costs of crime, x, are large, we presume that a policymaker will prefer to implement an outcome with a lower crime rate. Following these specifications, equilibrium (2.3) cannot be socially desirable, as it shows the highest probability of crime.[17] The other two equilibria show an equal and lower probability of crime, which is deemed socially preferable, but also diverging levels of inspection effort by corporate management and authorities.

Second, it is insightful to distinguish the occurrence of these equilibria from the pure strategy outcome. Consider Proposition 1 and allow its second precondition to fail, i.e. $c_M < (1-\beta)f_M - f_M^*$ applies instead. In other words, the monitoring costs c_M are lower than the gain from self-disclosure for M, so the manager is inclined to inspect the employee. Furthermore, given the strong incentive for "crime" of E (first precondition of Proposition 1), the pure strategy of A is not sufficient to turn the employee indifferent, and thus M plays "*inspect*" with positive probability. This leads to equilibrium (2.3). If we departed Proposition 1 by allowing its first precondition to fail instead, i.e. $b < (1-\beta)f_E$ holds due to a harsh punishment f_E of the employee, this would lead to equilibrium (2.2). Then, monitoring costs are still too high for M to pay off (second precondition of Proposition 1), so M does never report "*crime*". Given the lower private gains through non-compliance for E, however, it is sufficient that A (E) randomizes to make E (A) indifferent between the pure strategies.

6. Discussion

Having established our three-person inspection game framework, we can now discuss the impact of different liability regimes on criminal behavior and policing incentives.

6.1. A composite liability regime?

Our analysis identified two equilibria in mixed strategies which both showed an equally low crime rate, equilibrium (2.1) and equilibrium (2.2). Before we can discuss the design of a liability regime, we need to further distinguish between the two outcomes. Therefore, we seek to determine the total social costs of corporate crime and criminal prosecution. We specify three categories of social costs: costs of crime, inspection effort costs, and error costs. Table 1 compares the two equilibria regarding their social costs.

[17] Distinct from the other two equilibria, A always prefers the pure strategy "*inspect*" in (2.3). This can only be achieved by a higher probability of crime, as A is exactly indifferent in the other two outcomes.

Table 1
Social costs of crime and enforcement.

Equilibrium	Costs of crime	Inspection effort	Error costs
(2.1)	$\phi_E \cdot x$	$\phi_M^{2.1} c_M + \left(1-\phi_M^{2.1}\right)\left(\phi_E\left(1-\alpha-\beta\right)+\alpha\right)\phi_A^{2.1} c_A$	$\phi_E\left(1-\phi_M^{2.1}\right)\left(\beta+(1-\beta)(1-\phi_A^{2.1})\right) d$
(2.2)	$\phi_E \cdot x$	$\left(\phi_E\left(1-\alpha-\beta\right)+\alpha\right)\phi_A^{2.2} c_A$	$\phi_E\left(\beta+(1-\beta)(1-\phi_A^{2.2})\right) d$

The costs of crime that are to be expected are defined by the extent of damage to society and the probability that crime will occur. The second category of costs is incurred through criminal investigations and inspection efforts by M and A. The costs of errors result from decision errors in enforcement, i.e. wrongful acquittals, and society's aversion to such errors.[18] Comparing the two equilibria, we provide the relevant observations in Corollary 2.

Corollary 2. *Both equilibria differ only in inspection costs. The total social costs of equilibrium (2.2) are independent of player A's mixed strategy.*

Proof. Appendix A2.

Following these insights, a legal policy maker will have to compare only the expected inspection costs in order to choose between the two equilibria. Thus, equilibrium (2.1) is socially favorable if and only if the inequality

$$\phi_M^{2.1} c_M < c_A\left(\phi_E\left(1-\alpha-\beta\right)+\alpha\right)\left[\phi_A^{2.2}-\left(1-\phi_M^{2.1}\right)\phi_A^{2.1}\right] \tag{8}$$

holds. Generally, condition (8) is more likely to be satisfied if the cost advantage of managerial inspections compared to authorities is large. A legal policy maker will thus seek to implement equilibrium (2.1) if authorities face significantly higher investigation costs.

The choice between the two equilibria is also related to the choice of the liability regime. Equilibrium (2.1) requires a regime of composite liability, i.e. both the corporation and the employee have to be fined. Only this will induce the corporation to exert costly, but socially desirable inspection effort. Also, as long as fines cannot be transferred between the economic agents through contracting or civil litigation, only the risk of criminal punishment prevents the employee from playing the pure strategy "*crime*". In contrast to this liability regime, equilibrium (2.2) makes no such requirement. Theoretically, it would suffice only to have the wrongdoer criminally liable, i.e. the employee. No effort of M is required. As highlighted in Corollary 2, it is also noteworthy that total social costs of equilibrium (2.2) are independent of the chosen mixed strategy of A, and hence also of the level of punishment ultimately imposed on the employee. Although a higher punishment induces less effort by authorities, which saves costs, this is perfectly compensated by a higher level of error costs through wrongful acquittals and unprosecuted wrongdoers.[19]

As demonstrated by condition (8), our application of the three-player inspection game relates the gains of composite liability regimes to the relative cost disadvantage of authorities in investigating corporate crime. This is distinct from the standard line of argumentation for vicarious liability where the main reasoning for corporate liability lies with the insufficient wealth of the wrongdoer (see [23], p. 670). Our analysis also indicates that even under the idealized ability of the corporation to monitor its employee, the more complex composite liability regime may still not always be desirable for efficiency reasons.[20]

Furthermore, it needs to be emphasized that condition (8) is not exogenously given but is (to some extent) dependent on the legal policy variables f_E, f_M and f_M^*. In other words, what liability regime is desirable also depends on the ability of authorities to impose these punishments on the economic agents. Even if theoretically feasible, there can be upper limits on fines, fines may be reallocated, or an agent may even be judgment-proof. Moreover, punishing one agent may interfere with efficient incentives for the other agent. Such considerations are particularly relevant for the choice of a composite regime. For example, an increase in the punishment f_E for the employee will decrease the inspection effort of the manager and authorities, i.e. we find $\frac{\partial \phi_M^{2.1}}{\partial f_E} < 0$, and $\frac{\partial \phi_A^{2.2}}{\partial f_E} < 0$. Reconsidering (8), there is a direct cost-saving effect on both sides of the inequality, but also a countervailing cost increase for equilibrium (2.1) as less monitoring by M implies a higher probability of involvement of the high-cost inspector A.[21] Turning to the sanctions f_M for the corporation, a higher punishment f_M will c.p. shift inspection effort from authorities to the manager, i.e. we find $\frac{\partial \phi_M^{2.1}}{\partial f_M} > 0$, and $\frac{\partial \phi_A^{2.1}}{\partial f_M} < 0$. While this implies an increase on both sides of the inequality (8), this effect will be amplified by the relative cost-disadvantage c_A of player A. In other words, this inequality is more likely to hold when sanctions f_M on the manager are high. For such a composite liability regime to work, however, a sufficient sentence discount has to be offered for truthful reporting of wrongdoers by player M. As $\frac{\partial \phi_A^{2.1}}{\partial f_M^*} > 0$ and $\frac{\partial \phi_M^{2.1}}{\partial f_M^*} < 0$ apply, offering a lower punishment induces more inspection effort by corporate management.[22] Following our analysis, a composite liability regime requires a sufficiently high punishment for the corporation, a high discount for reporting of wrongdoing, and a limited[23] punishment for the employee. This is because a high punishment for the employee degrades incentives for the management to exert inspection effort.

It is remarkable that social costs for equilibrium (2.2), however, are not contingent on the strategic choice of authorities, and thus not affected by the level of punishments. So, while the social desirability of the composite liability systems relies on the correct specification of fines to incentivize M, the criminal liability regime of equilibrium (2.2) is not prone to such specification errors. Furthermore, only equilibrium (2.2) can maintain the lowest crime rate when we consider a deviation to the basic game where costly inspection effort by M does no longer prevent erroneous prosecution by authorities.[24]

6.2. Reallocation of punishments

In the proposed inspection game, payoffs are assumed to be final, that is, not subject to bargaining or reallocation through contracts or civil litigation. While there are certainly legal and economic limitations on transferring sanctions from one player to the other, such reallocation has received considerable interest in the literature (see, e.g., [1,2]). In

[18] Be reminded that due to the assumption that authorities only impose a punishment after investigations, and investigations always reveal the true behavior of the employee, only wrongful acquittals are a concern in our compliance game.

[19] Using *Proposition 2.2*, we find $\frac{\partial \phi_A^{2.2}}{\partial f_E} < 0$.

[20] Note that in addition to its cost advantage, inspections by M perfectly avoid wrongful accusations and wrongful acquittals. Both type of errors can be expected to reduce the deterrence of crime.

[21] Also be reminded that a too high f_E will c.p. violate the requirements for Proposition 2.1 to hold.

[22] Also note that a high sentence discount, $f_M - f_M^*$, favors the requirements of Proposition 2.1.

[23] Be reminded that a minimum punishment is required to prevent E from always playing "*crime*".

[24] See Appendix 5.

the following, we want to briefly discuss how a contractual transfer of fines would affect the outcome of the game.

Assume that the corporation has all the bargaining power in the labor contract. It would then offer a profit-maximizing contract to the employee, and this contract is to (i) incentivize E not to commit a crime[25] and (ii) to participate in the contract, nonetheless. Contract participation is not a binding constraint, however, as wrongful convictions that require contractual compensation cannot occur in our basic inspection game. Thus, the corporation could impose a contractual sanction on E and theoretically require total compensation for its damage, i.e. the loss through crime and the received punishment. Clearly, this contractual sanction only applies when E is caught, so the employee receives a higher effective punishment $f_E{}'$, with $f_E{}' > f_E$. As established by Proposition 2, the crime rate is independent of the employee's punishment (denoted f_E previously). Accordingly, any contractual reallocation has no effect on crime in equilibrium. However, the behavior of the manager changes when punishments can be passed on to the employee: the management is then indifferent between inspecting the agent or not inspecting, if only $-c_M = -\mu_M\left(\beta+(1-\beta)(1-\phi_A)\right)\gamma x$ applies. Using $\mu_M = \phi_E$ and solving for ϕ_A, we find former condition (3) modified to $\phi_A = \frac{\phi_E\gamma x - c_M}{\phi_E(1-\beta)\gamma x}$.[26] A's mixed strategy is apparently no longer responsive to the level of punishment imposed on the corporation. As $\frac{\partial\phi_A}{\partial\gamma x} > 0$, the contractual reallocation incentivizes authorities to intensify investigative effort when the damage to the corporation increases. Moreover, condition (5) indicates ($\frac{\partial\phi_M}{\partial f_E} < 0$) that M would further reduce investigative effort as $f_E \gtrless f_E$ applies. While such contractual reallocation shows no impact on the deterrence of crime, it evidently degrades the efficient use of the policy variables f_E and f_M in a composite liability regime.

7. Extensions

In the following, we will extend our compliance model to study some other applications of criminal liability regimes that have been previously addressed in the literature. For the sake of brevity, we just explain the basic changes and outcomes and leave theoretical reasoning mostly to the appendix.

7.1. Prevention of crime and negligence

Distinct from strict liability, also negligence-based liability rules can induce optimal behavior and deterrence. Under a negligence standard, "firms would be liable only when they failed to make an adequate investment in monitoring their employees" (Hamdani and Klement, 2008, p. 301). Courts will then usually have to determine what level of investment is 'adequate' in this regard. However, the inspection game in its proposed form is unsuitable to study negligence-based liability, as inspections by corporate management in our setup imply reporting non-compliance to authorities, but no measures to prevent the opportunities for criminal behavior within the organization. In the words of Arlen [7, p. 144] we have to adapt our model to consider also the ex-ante deterrence of crime.

Consider the proposed basic game (see Fig. 1) as a proper subgame to a larger dynamic game which contains one additional stage: assume that the larger game begins with M who decides whether to implement preventive organizational measures against crime, which we designate "*prevention*", or not. Assume that choosing "*prevention*" produces additional costs p, but prevents all criminal opportunity within the organization with certainty. Alternatively, one can interpret "prevention" as adhering to the negligence standard and this perfectly insulates M from any punishment. In this interpretation, p describes the total costs of ex-ante deterrence, including prevention costs and remaining (financial) damage through crime for the corporation.[27] Only when M decides against preventive measures does the basic compliance-subgame unfold. Its equilibria have been established above.

Let us first consider a scenario where the subgame yields *equilibrium (2.1)* and a composite liability regime is applied. In this fully mixed strategy outcome, M is indifferent between her pure strategies "*inspect*" and "*not*", and this would yield the equilibrium payoff $-\left(\gamma x + f_M^*\right)\phi_E - c_M$. Consequently, the manager will choose preventive measures at the beginning of the modified game, if and only if

$$p < \left(\gamma x + f_M^*\right)\phi_E + c_M \tag{9}$$

Evidently, M is more inclined to prevent crime ex-ante if the expected damage to the firm γx is high, the costs of ex-post inspection effort c_M are significant, or if the probability of crime ϕ_E is high. Surprisingly, the decision *against* preventive measures becomes more favorable if f_M^* is low. In other words, if the firm can expect a punishment discount on reporting the crime truthfully ex-post, it is less willing to undertake costly prevention ex-ante. This reveals a relevant caveat of the composite liability regime. It relies on a significant punishment discount $(f_M - f_M^*)$ for the firm in order to incentivize desirable monitoring effort, but this reduces incentives for crime prevention. This finding resembles the analysis of Arlen [7, p. 145] who argues that the state "cannot simultaneously induce optimal prevention and policing".

Let us now consider the second scenario where the compliance subgame yields *equilibrium (2.2)*. In this case, the firm expects the payoff $-\phi_E\left(\gamma x + (1-\beta)\phi_A^{2.2} f_M\right)$. Consequently, M chooses "*prevention*" at the beginning of the game if

$$p < \phi_E\left(\gamma x + (1-\beta)\phi_A^{2.2} f_M\right). \tag{10}$$

Generally, preventive measures are more favorable the higher the expected damage through crime and the higher the expected punishment. Due to $\frac{\partial\phi_A^{2.2}}{\partial f_E} < 0$, inequality (10) also shows that an increase in the punishment of the employee f_E leads to less expected punishment for the corporation. Again, the incentive for crime prevention by M is diminished. As established in Section 6.1, the total costs of crime and prosecution are not dependent on the level of enforcement by authorities $\phi_A^{2.2}$ under Proposition 2.2. Making use of this finding, a lower sanction f_E could increase $\phi_A^{2.2}$ and thus incentives for preventive measures according to (10) without additional social costs in the basic compliance sub-game.

7.2. 'Blind justice'

Law and economics scholars have repeatedly emphasized that authorities have significant difficulties investigating corporate crime and that the legal policy maker should make better use of corporate management (see, e.g., [2]). We have considered a three-person inspection game where corporate management and public authorities are both able to uncover a criminal action, although authorities suffer from a disadvantage in investigation costs and their limited ability to select potential cases on their own. Now consider a more pronounced disadvantage for authorities: they can still enforce criminal law on all economic agents but are completely unable to uncover the truth without the help of the corporation. For our setup, authorities are no

[25] In our model, the firm faces a damage of γx due to the criminal behavior of its employee. Of course, there are scenarios where a firm may benefit from the corporate crime (i.e. $\gamma < 0$ applies). It could then even incentivize E to commit the crime by offering compensation for the punishment f_E, given that the corporate gains are sufficiently high. However, this partly dissolves the studied inspector–inspectee relationship.

[26] Note that the mixed strategy requires (i) $\phi_E\gamma x > c_M$ and (ii) $c_M/\left(\phi_E\gamma x\right) > \beta$.

[27] While we model "*prevention*" as a discrete strategy for simplicity, it is easy to image that these specific prevention costs p are the result of a manager who considers the optimal level of prevention effort, given that liability costs and prevention costs are continuous functions of this effort.

longer an inspector in the tradition of inspection games, but a mere enforcer who only chooses between the strategies "*punish*" and "*not punish*". Nevertheless, be reminded that A still cares about decision errors. Furthermore, a corporation that exerts inspection effort and thus reports non-compliance to authorities still receives a correct decision by A. In this modified game, there are two equilibria in pure strategies, although only one equilibrium is credible.[28]

Proposition 3. *(i) Strategies {"crime"; "notinspect"; "punish"} and beliefs $\mu_M = 1$ and $\mu_A = 1$ constitute a PBE if $b > (1-\alpha-\beta)f_E$ and $c_M \geq (1-\beta)f_M - f_M^*$ apply. (ii) Strategies {"notcrime"; "inspect"; "punish"} and beliefs $\mu_M = 0$ and $\mu_A = [0,1]$ constitute a PBE if $c_M \leq \alpha f_M$ apply. This PBE is not sequentially rational.*

Proof. Appendix 2.

Evidently, the impact of authority's inability to verify corporate crime on the pure strategy outcome is limited: the requirements for the inefficient strategy set {"crime"; "notinspect"; "punish"} are only marginally changed compared to the basic model. A second PBE must be discarded, as it is never credible that authorities will punish an employee who chooses to comply with the rules.

Proposition 4. *The following mixed strategy equilibria exist:*
(4.1) $\phi_E^{4.1} = \frac{\alpha}{1-\beta+\alpha}$, $\phi_M^{4.1} = \frac{bf_M\left[\phi_E^*(1-\alpha-\beta)+\alpha\right]-(\phi_E^* f_M^* + c_M)(1-\alpha-\beta)f_E}{f_E f_M\left[\phi_E^*(1-\alpha-\beta)+\alpha\right]-(\phi_E^* f_M^* + c_M)(1-\alpha-\beta)f_E}$ *and* $\phi_A^{4.1} = \frac{\phi_E^* f_M^* + c_M}{\left(\phi_E^*(1-\alpha-\beta)+\alpha\right)f_M}$, *given beliefs $\mu_M = \phi_E$ and $\mu_A = \frac{1}{2}$. This requires* $c_M < \phi_E\left[(1-\alpha-\beta)f_M - f_M^*\right] + \alpha f_M$.
(4.2) $\phi_E^{4.2} = \frac{\alpha}{1-\beta+\alpha}$, $\phi_M^{4.2} = 0$ *and* $\phi_A^{4.2} = \frac{b}{(1-\alpha-\beta)f_E}$, *given beliefs $\mu_M = \phi_E$ and $\mu_A = \frac{1}{2}$. This requires the conditions $b\langle(1-\alpha-\beta)f_E$ and $c_M\rangle\phi_E\left[(1-\alpha-\beta)f_M - f_M^*\right] + \alpha f_M$ hold.*
(4.3) $\phi_E^{4.3} = \frac{c_M - \alpha f_M}{(1-\alpha-\beta)f_M - f_M^*}$, $\phi_M^{4.3} = \frac{b-(1-\alpha-\beta)f_E}{(\alpha+\beta)f_E}$ *and* $\phi_A^{4.3} = 1$, *given beliefs $\mu_M = \phi_E$ and $\mu_A > \frac{1}{2}$. This requires the conditions $b > (1-\alpha-\beta)f_E$, $c_M < (1-\beta)f_M - f_M^*$, $c_M > \alpha f_M$ and $\phi_E^{4.3} > \phi_E^{4.1}$ to hold.*

Proof. Appendix.2

The equilibrium in mixed strategies shows several interesting features: first, two equilibria, (4.1) and (4.2), again show a lower crime rate than the third equilibrium. Second, the probability that 'blind' authorities impose a punishment is lower compared to the probability of investigations in the basic game if M also exerts inspection effort. If M remains inactive, however, the opposite applies, and the probability of punishment is higher. The reason for this variation is the added risk of wrongful convictions and false allegations due to A's inability to verify the true behavior of E. One could thus conclude that a composite liability regime that induces M to exert control effort is even more favorable when the verification skills of authorities are weak.

It is easy to see that a different policy instrument to avoid the risk of wrongful convictions proves to be rather ambiguous: imposing a higher standard of proof on criminal procedures. In the stylized setup of our inspection game, a standard of proof is the required ex-post belief by authorities to punish a wrongdoer. Given that we assumed error costs d to be equal both for wrongful convictions and acquittals, the standard of proof is simply $\mu_A \geq 1/2$.[29] Society's preferences can require a higher decision standard, such as, for example, $\mu_A \geq 0.75$. Following Tsur [57] and Christmann and Kirstein [58], such a threshold could be interpreted as society being indifferent between three wrongful acquittals and one wrongful conviction. However, this implies that the ex-post belief of authorities would have to be increased to $\mu_A = 0.75$ if A is to be indifferent between "*punish*" and "*not punish*". Given that only E can turn A indifferent, this requires a *higher* rate of crime. In other words, raising the standard of proof reduces decision errors through wrongful convictions but comes at the cost of increased corporate crime.

7.3. Whistleblowers

Distinct from corporate management, whistleblowers could also provide valuable information to prosecutors and assist authorities to punish wrongdoers. A scenario where a whistleblower can observe the potentially criminal behavior of fellow employees or managers at no cost, however, lies outside the scope of the inspection game and this paper: the equilibrium would fully depend on whether the whistleblower is willing to report observed criminal behavior when facing potential retaliation by the firm. But what if some vigilance and attentiveness of the whistleblower is required to observe such criminal behavior of others?

Consider the extensive form of the basic compliance game (see Fig. 1) but let player M be designated as the whistleblower within the firm. Assume that whistleblower M can choose between the strategies "*inspect*" and "*not*". The strategy "*inspect*" then implies that the whistleblower is vigilant and attentive, and choosing "*inspect*" leads to a perfect revelation of E's true behavior. Furthermore, let M be motivated by her own personal sense of integrity and desire to have wrongdoers convicted. In the world of our model, M receives a private gain w when criminal behavior by E is revealed and punished, i.e. either M or A play "*inspect*". When she chooses the strategy "*inspect*" herself, she incurs effort costs c_M. Moreover, M also expects a loss r through retaliation by the corporation if a crime is reported. For the whistleblower, the game yields the following outcomes: (i) the payoff $w - c_M - r$ if she exerts effort and detects a crime, (ii) the payoff $-c_M$ if she exerts effort, no crime is detected and the game ends nonetheless,[30] (iii) the payoff $(1-\beta)w$ if criminal behavior is revealed and punished without her involvement, and (iv) a payoff of zero otherwise. This setup again shows the dilemma of the inspection game: the whistleblower wants crime to be punished but would save private effort if there were no crime. In addition, M will never learn whether a crime occurred if neither player chooses "*inspect*". All payoffs of the other two players are unchanged. See *Appendix A4* for the whistleblower's payoff matrix. Having introduced the whistleblower to the compliance game, we can derive the following two implications for the previously discussed equilibria.

First, excessive retaliation will rule out any desirable control by whistleblowers. More specifically, the whistleblower is only a relevant player if $w - c_M - r > 0$ holds. Otherwise, "*not*" is always the dominant strategy for M. This means that M's incentive to report observed wrongdoing must exceed potential effort costs and retaliation. Moreover, let us consider the pure strategy equilibrium {"crime"; "not"; "inspect"}. As noted above, this is not a desirable outcome, as criminal behavior always occurs. For M to choose the pure strategy "*not*" over blowing the whistle, this requires the condition $w - c_M - r < (1-\beta)w$ to hold. This is reminiscent of a freeriding dilemma, i.e. the whistleblower will prefer to stay silent if effort costs c_M or retaliation r are significant or if the probability of inspections by authorities, $(1-\beta)$, is already sufficiently high. It is straightforward to see that a distinct reduction in retaliation costs r, for example due to a whistleblower protection program, can potentially eliminate this undesirable outcome (given effort costs are not too high, i.e. $\beta w > c_M$ applies).

[28] See Appendix 3 for the normal form of the amended game.

[29] Authorities will choose "*punish*" over "*not punish*" if $-(1-\mu_A)d \geq -\mu_A d$ holds, which yields $\mu_A \geq 1/2$.

[30] For simplicity, we keep to the game tree of the basic game, thus the game always ends when M exerts effort. At least for the case of the innocent E, this appears to assume away the problem of fraudulent charges (by 'nature') despite a vigilant M. Nevertheless, we are concerned with the whistleblower's incentives and behavior here, and M's payoffs would not be affected if we allowed the game to continue when M inspected an innocent E.

Second, more whistleblower protection results in less whistleblower activity. To see this, we turn to the derivation of the mixed strategy outcome. Given the considerations above, this clearly requires $w - c_M - r > (1-\beta)w$. Then, W can be made indifferent between her pure strategies if $\phi_E(w-r) - c_M = \phi_E\phi_A(1-\beta)w$ apply. Solving for ϕ_A, we thus determine the probability of inspections by authority ϕ_A as[31]

$$\phi_A = \frac{\phi_E(w-r) - c_M}{\phi_E(1-\beta)w} \tag{11}$$

Reconsidering that a whistleblower protection program should result in a reduction of r, this implies a higher probability of inspections by authorities ($\frac{\partial \phi_A}{\partial r} < 0$). Using $\frac{\partial \phi_M}{\partial \phi_A} < 0$ from Proposition 2, more inspections by A induce *less* whistleblowing in equilibrium. At the same time, the crime rate is unaffected by enhanced whistleblower protection. Although we deduce from the three-player inspection game that higher whistleblower protection should lead to more investigations by prosecutors, this results in the unintended adverse effect of less whistleblowing and no effect on crime. Our inspection game setup thus offers another explanation for the experimental evidence of Mechtenberg et al. [35]. The authors also find no effect of whistleblower protection on crime, but in their setup, under the assumption that whistleblowers can observe criminal behavior at *no cost*, a higher whistleblower activity is totally compensated by less inclination of authorities to investigate reports.

8. Conclusions

In this paper, we modeled corporate compliance as a three-person inspection game in order to analyze the impact of different liability regimes on rule breaking behavior by the firm's agents, policing effort by corporate management and enforcement effort by authorities. In contrast to the management, public authorities are often considered to be at a disadvantage when prosecuting corporate crime. We integrated this caveat into our model by higher inspection costs for authorities and that, outside voluntary self-disclosure by the firm, a crime may go unnoticed or false accusations are reported with positive probability.

Our inspection game featured two equilibria in mixed strategy that showed a positive, but relatively low probability of crime. While in the first equilibrium, all players randomize and both the manager and authorities inspect the employee with positive probability, in the second equilibrium only authorities show inspection effort. Which equilibrium is socially more desirable with regard to social costs depends largely on the relative difference in monitoring costs between the corporation and authorities. If the first outcome is indeed preferable, a composite liability regime must be implemented: it requires a sufficiently high punishment for the corporation, a high discount for reporting of wrongdoing, and a limited punishment for the employee. If the second outcome is more favorable, only the punishment for the wrongdoer is required. Also, its implementation is less prone to specification errors as its social costs are independent of the behavior of the two inspectors. This becomes even more pronounced when punishments can be reallocated among the economic agents by contracts or civil litigation, as this further interferes with the use of the policy variables of the composite liability regime. This second outcome is also more robust in a scenario where ungrounded, erroneous prosecution can occur despite applied internal control measures by the corporation. However, whenever the verification skills of authorities are weak, even to the degree of "blind justice", a composite liability regime that can induce self-disclosure by firms clearly becomes preferable.

Our model also offers further interesting applications. Considering a negligence-based liability regime, we find that incentives for the ex-ante prevention of crime interfere with optimal incentives for policing activity and self-disclosure by the firm. With regard to whistleblower protection programs, we find that they induce more investigations by prosecutors, but, quite surprisingly, less whistleblowing activity and no effect on crime.

[31] Note that for (11) to be a mixed probability, it is required that (i) $\phi_E(w-r) > c_M$, and (ii) $\phi_E(w-r) - c_M < \phi_E(1-\beta)w$, which simplifies to $c_M > \phi_E(\beta w - r)$.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.dajour.2024.100431.

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